

A National Airspace System Reference Architecture as a System of Systems

Tracing Operational Intent from Enterprise Objectives to Aircraft
Trajectory

ISD 503 Capstone Project Report – Interim Report #1

Submitted to:

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Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section 5.0 Impact/Financial Benefits), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report.

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1 Introduction and Current State

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

My project aims to build a reference MBSE architecture for how airspace intent is managed across the NAS — spanning the organizations that define and enforce rules, the domains managing airspace resources, the systems supporting flight operations, and the onboard systems that turn intent into executable trajectory and guidance behavior.

In his 2005 paper, DeLaurentis states the following DeLaurentis [8]:

“PROBLEMS of significant complexity are facing decision-makers within government and industry, and the moniker of ‘system-of-systems’ (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the ‘stovepipe’ mode – their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems, exemplified in this paper by future national transportation system realizations. A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time. And this need is urgent, for the SoS type almost always involves decisions that commit large amounts of money, for which ultimate failure or success carries heavy consequences over several generations, and that impact large segments of the public. The costs associated with a chosen path may still be high, but the intent is to maximize the probability that these costs result in good consequences rather than bad.”

While this reference is from 2005, it is still relevant today. The NAS is a complex system of systems, and the decisions made by the various stakeholders have significant consequences for the safety, efficiency, and sustainability of the air transportation system.

Airline companies invest significant time and money in constructing operational plans that maximize revenue; however, on the day of operation, unpredictable events – bad weather, aircraft malfunctions, crew absenteeism – can substantially alter that plan and the revenue objectives behind it Castro [5]. To manage this, airlines maintain an Airline Operations Control Center (AOCC), a department of experts specialized in disruption management across aircraft, crew, and passenger recovery. The AOCC monitors operations and makes real-time decisions – re-routing flights, rescheduling crews, rebooking passengers – intended to minimize the impact of a disruption on the airline’s schedule and revenue objectives.

The literature does not converge on a single, standardized definition of the “cost” the AOCC is minimizing, and the values it does quantify do not always hold up under scrutiny. Delay cost is commonly modeled in the optimization literature as a linear function of delay minutes, but Hu et al. [21] note this assumption is inconsistent with airline practice, where delay cost is in fact a nonlinear function of delay time – and that, faced with this complexity, airlines tend to prioritize the speed of a recovery decision over the mathematical optimality of the resulting solution. This preference for tractable, fast decisions over globally optimal ones shows up structurally as well

as behaviorally: a systematic review of the Aircraft Recovery Problem finds that studies – and, by extension, the AOCC processes they model – rarely minimize delay, cancellations, and aircraft swaps simultaneously, instead treating aircraft, crew, and passenger recovery as separable sub-problems solved sequentially and handed off in turn Santana et al. [41]. Each stage can produce a solution that is locally optimal yet globally suboptimal, or even infeasible, once the next stage’s constraints are applied – a concrete, literature-documented instance of the myopic-optimization pattern this capstone investigates.

That pattern’s costs and benefits are visible even in work that improves on the status quo. Kontodimou et al. [26]’s predictive turnaround-buffer allocation model achieves a 37.1% reduction in propagated delay relative to a zero-buffer baseline, but it reaches that result by holding aircraft routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as upstream constraints, deliberately excluding them from the optimization it performs. The gain is real within that bounded scope, but the framing illustrates the exact risk this capstone is concerned with: a locally-optimized decision layer can report a strong local benefit while externalizing tradeoffs onto adjacent decision layers that go unmeasured by that layer’s own cost model. The same pattern appears on the human side of the AOCC: Turhan, Gzara, and Elhedhli [47] documents that when dispatcher workload exceeds capacity, dispatchers “may resort to conservative approaches, which can have financial implications for airlines” – a workload-driven cost that rarely appears in the delay, cancellation, and swap cost models discussed above. Hassan, Santos, and Vink [16] frames this more broadly as a persistent gap between the disruption-management optimization literature and the reality of AOCC practice.

In this paper, I explore the value tradeoffs that aviation stakeholders experience when disruptions occur and recovery solutions are employed with various levels of myopia. When a dispatcher or AOCC team member makes a decision, they are often focused on the immediate impact of that decision – e.g., minimizing delay for a specific flight – without visibility into the broader implications for the airline’s overall operations or the NAS as a whole. Cost models used today typically capture the factors that are easiest to quantify, such as fuel consumption and crew costs, while factors that are harder to quantify – maintenance cost, passenger satisfaction, dispatcher/controller workload, environmental impact – are more often overlooked. This can lead to suboptimal outcomes: increased congestion in certain airspace sectors, missed opportunities for more efficient routing, or decreased overall profits driven by customer dissatisfaction. By modeling the NAS as a system of systems and analyzing the propagation of operational intent, I aim to identify strategies for improving decision-making and reducing the negative impacts of disruptions on the air transportation system.

1.1 Overview of the Project

This project exists within the industry and operating environment of U.S. commercial air transportation – specifically, the National Airspace System (NAS): the combination of airspace, airports, aircraft, air navigation facilities, and the regulatory and operational organizations that use them to move passengers and cargo. The NAS is not one system but, as introduced above, a System of Systems: it is jointly owned and operated by organizations with distinct, only partially overlapping objectives – airlines, airports, the FAA and its Air Traffic Control (ATC) facilities, and the aircraft and crews that execute the plans those organizations produce. No single one of these organizations has authority over the whole; each manages its own slice of the problem and hands intent – a schedule, a flight plan, a clearance, a trajectory – to the next.

The social context surrounding this environment is significant on several axes at once. It is safety-critical: air traffic control exists principally to guarantee separation between aircraft, and

the flight crew’s foremost authority is to deviate from any assigned plan to preserve that safety margin. It is economically significant: a single major carrier’s irregular operations – weather, mechanical failures, and crew unavailability – can exceed \$440 million per year in lost revenue, crew overtime, and passenger accommodation costs, and a single severe-weather event (the “Blizzard of ’96”) cost the U.S. airline industry an estimated \$50-100 million in one week Clarke [7]. It depends on specialized human operators – dispatchers and controllers – whose workload is itself a scarce, unpriced resource: dispatchers facing workload beyond their capacity “may resort to conservative approaches, which can have financial implications for airlines” Turhan, Gzara, and Elhedhli [47], and controller workload is a hard operational ceiling on how much traffic a sector can safely absorb. And it has environmental and community consequences – fuel burn, emissions, and aircraft noise – that are borne by parties (future generations, communities near airports) who are not party to the operational decisions that produce them. Section 1.3 develops this stakeholder landscape in full; the point to establish here is that the NAS’s operating environment already contains, in miniature, the tension this project investigates: safety, economics, human workload, and environmental cost are managed by different organizations, on different timescales, using different and sometimes incompatible definitions of “good.”

I selected this topic for two reasons. First, the NAS is one of the most mature, best-documented examples of a real, operating System of Systems available to a systems-engineering capstone: the FAA and its international counterpart, EUROCONTROL/SESAR, publish detailed public planning and architecture documents (e.g., Federal Aviation Administration [13] and SESAR Joint Undertaking [44]), and decades of operations-research and disruption-management literature document how its constituent organizations actually make decisions. That makes it possible to build a grounded architecture rather than a purely hypothetical one. Second, and more specifically, this project began as a narrower trajectory-optimization problem and was deliberately broadened into an architecture of the whole intent-propagation chain (see Scope and the project’s decision log, D-001) because that framing better exercises the skills an ISD capstone is meant to demonstrate – structure, interfaces, responsibility, and traceability – while still leaving room for an optimization case study that shows why those skills matter operationally.

The potential benefit of this work is a reference architecture that makes cross-stakeholder trade-offs visible *before* they are made, not after. As the literature reviewed in Section 1.2 shows repeatedly, today’s decision-support tools and human decision processes are organized around one organization’s cost function at a time; the costs that decision externalizes onto another stakeholder are usually invisible to the tool or person making it. An architecture that traces how operational intent and information actually flow – from an enterprise objective, through airline and ATC planning, to the trajectory an aircraft actually flies – gives a systems engineer, and eventually a decision-support capability built on that architecture, a place to see those externalities before they are absorbed as unplanned cost by someone else.

Six questions motivate the work and give it a testable shape (see also the project’s D-004 decision record):

- *How should the NAS be decomposed into constituent systems and Systems-of-Systems?* – the architecture is only useful if its boundaries reflect real authority and information-ownership lines, not an arbitrary catalog of aviation objects.
- *Where do authority, responsibility, ownership, and lifecycle boundaries occur?* – these are exactly the transitions (e.g., dispatcher to controller, controller to controller, controller to flight crew) where intent can be lost, delayed, or reinterpreted.
- *How does operational intent propagate from enterprise objectives to physical aircraft behavior?*

– this is the project’s organizing throughline (Strategic objective → mission plan → flight plan → ATC constraints → trajectory negotiation → FMS intent → guidance commands → aircraft motion) and the spine the architecture is built around.

- *How do stakeholder definitions of "optimal" differ?* – Section 1.3 and the project’s objective/cost ontology show that "optimal" means something different, and sometimes incompatible, to an airline, a controller, a passenger, and a community near an airport.
- *When does subsystem optimization produce undesirable System-of-Systems outcomes?* – this is the myopic-optimization pattern documented throughout Section 1.2, and the specific failure mode this project is trying to expose rather than hide.
- *How can an MBSE architecture support analysis of those tradeoffs?* – this is the project’s methodological bet: that a traceable model, not a one-off analysis, is what lets a tradeoff surfaced in one scenario generalize to the next.

Taken together, these questions state the problem this project addresses: NAS decision-making is fragmented across specialized organizational silos – airline planning, day-of-operations control, air traffic control, airport operations – each of which optimizes against its own, usually well-defined cost model, while the couplings *between* those silos are rarely modeled explicitly and are usually only discovered after a disruption forces them into view. That matters because, as Section 1.2 documents, this is not a hypothetical risk: it is the specific, recurring pattern the disruption-management and airline- scheduling literatures describe when they find that sequential, decomposed optimization is common in practice and produces solutions that are locally optimal but globally suboptimal or infeasible once handed to the next stage. Building a reference architecture that makes that propagation chain explicit is the first step toward analyzing, rather than merely enduring, the tradeoffs it creates.

1.2 Review of Current State Design / Literature Review

As of this report, 52 sources have been collected and triaged into a source register (bibliography entry, summary, and 0-5 relevance rating for each); roughly a quarter have received a full field-by-field deep annotation, with the remainder prioritized for the literature-review milestone that follows this interim report. What follows synthesizes the current state of four literature tracks: (1) benchmark documents describing how NAS-scale architecture is actually planned today, (2) airline planning and scheduling, (3) day-of-operations and turnaround coordination, and (4) operations control, dispatch, and disruption management – closing with the gap in that literature this project addresses.

Benchmark: how NAS-scale architecture is planned today

The closest thing to a "best-in-class" comparison for this project’s own deliverable – a NAS reference architecture – is the pair of national/ regional ATM modernization roadmaps the FAA and its European counterpart publish. The FAA’s *NAS Enterprise Architecture: Infrastructure Roadmaps* lays out a 17-year, domain-by-domain (Aircraft, Airport, Airspace & Procedures, Automation, Communication, Navigation, Safety, Surveillance, Weather, and others) investment plan from an "As-Is" to a "To-Be" NAS Federal Aviation Administration [13]. Its European counterpart, the SESAR Joint Undertaking’s *European ATM Master Plan*, sets out a comparable "Digital European Sky" vision built around trajectory-based operations (TBO), a service-oriented delivery model, and an explicit target enterprise architecture diagram SESAR Joint Undertaking [44]. Both are

authoritative and current, but neither publishes the MBSE derivation behind its architecture – they present the resulting roadmap, not a traceable model connecting operational concept to system architecture.

That derivation step does have a documented precedent, just not from the NAS: the EU-China GreAT (Greener Air Traffic Operations) research project publishes a three-volume progression that starts from a current-state TBO concept baseline Finke et al. [14], uses an explicit MBSE methodology to derive a gate-to-gate operational and system architecture from that concept Hu et al. [19], and extends the result down to an avionics-level architecture. This is a rare, concrete example of exactly the operational-concept-to-architecture move this capstone’s own methodology (SysML in Cameo, informed by MagicGrid and selected UAF/DoDAF concepts – decision D-003) needs to make for the NAS, and the closest available benchmark for judging whether this project’s own architecture is comparably rigorous. A second, smaller-scale precedent for the same ontology-first, MBSE-second move applied to an aviation System of Systems is Sinha Roy et al. [46]’s architecture framework for Urban Air Mobility, and Liu et al. [27] demonstrates a full, requirements-traceable MBSE pipeline (metamodel, architecture model library, and formal verification against ICAO requirements) for an ATM system in a different regulatory context – together these establish that MBSE-derived, requirements-traceable ATM architecture is an active, current methodology, not a hypothetical one, even though no published example does it for the U.S. NAS specifically.

Airline planning and scheduling

The airline-planning literature converges on one finding relevant to this project regardless of which sub-problem is being studied: Eltoukhy, Chan, and Chung [10]’s review of over 100 papers organizes airline schedule planning into four sequential sub-problems – flight scheduling, fleet assignment, aircraft-maintenance routing, and crew scheduling – solved one after another, and finds that this sequential decomposition is the industry norm even though it is provably suboptimal relative to integrated formulations. Lohatepanont and Barnhart [28] demonstrate a concrete integrated schedule-design-and-fleet-assignment model, using real airline data, that outperforms the sequential approach precisely because it does not treat fleet assignment as a fixed input to be optimized around after schedule design is already frozen. Xu, Wandelt, and Sun [49] extends this picture with explicit mathematical formulations for thirteen representative scheduling models and highlights robust-scheduling variants that explicitly price in disruption risk – a step toward internalizing, rather than externalizing, the very costs that later disruption-management research (below) has to absorb after the fact. The throughline for this project is that the sequential, compartmentalized planning process is not a strawman: it is the documented, still-common industry practice, and its cost is that decisions made at one stage (e.g., fleet assignment) become fixed constraints the next stage (e.g., crew scheduling) cannot renegotiate even when it would be globally beneficial to do so.

Day-of-operations and turnaround

Turnaround is the clearest example in this literature of a process that is technically local (one aircraft, one gate, one set of ground crews) but whose information dependencies are not: Schmidt [42]’s review of turnaround operations and simulation approaches documents how airport capacity pressure and schedule disruption both concentrate at the turnaround, and surveys the agent-based, discrete-event, and cellular-automaton simulation frameworks used to study it. EUROCONTROL’s current *Specification for Airport Collaborative Decision Making* EUROCONTROL [11] is the operational answer to that information problem: rather than optimizing turnaround locally, it defines a shared “milestone approach” – explicit target, estimated, and actual times for each turnaround

event – that airports, airlines, ground handlers, and the Network Manager all update and read from a common information model. A-CDM is, in effect, an existing, deployed example of the kind of information-sharing architecture this capstone is trying to generalize across the rest of the NAS: it does not eliminate each stakeholder’s local objective, but it makes the information those objectives depend on visible across organizational boundaries instead of trapped inside one of them.

Operations control, dispatch, and disruption management

This is the deepest and most directly relevant of the four tracks, and its findings are woven through Section 1’s framing above; this subsection adds the pieces not already covered there. Clarke [7]’s foundational 1998 survey of the Airline Operations Control Center (AOCC) – still cited as the origin point of this literature by later reviews such as Hassan, Santos, and Vink [16] – establishes the AOCC’s basic structure (a central control center supported by maintenance- and station-level control centers) and its guiding decision rule: return to the planned schedule as fast as possible, subject to crew and maintenance feasibility, largely independent of the trajectory-level fuel or routing consequences of how that return is achieved. Hassan, Santos, and Vink [16]’s more recent review formalizes why that rule persists: aircraft, crew, and passenger recovery are treated as separable decision domains not because they are independent in reality, but because integrated (multi-resource) recovery is harder to solve and the literature that could support it does not match AOCC practice – Hassan, Santos, and Vink describe this explicitly as a gap between published research and the reality of AOCC decision-making. Bouarfa, Müller, and Blom [4] and its originating thesis Castro [5] offer a partial counterexample: the MASDIMA multi-agent architecture assigns each resource (aircraft, crew, passengers) its own agent with an explicit utility function and negotiates a joint solution through a supervisor agent – a working, implemented demonstration that local objectives *can* be reconciled algorithmically rather than left to sequential handoff, though Bouarfa, Müller, and Blom’s own evaluation finds that automating those roles concentrates workload and judgment onto the remaining human supervisor rather than eliminating the need for it.

The pattern above is not limited to human decision processes; it recurs even in mathematically rigorous work. Santana et al. [41]’s systematic review of the Aircraft Recovery Problem finds that studies (and, by extension, the AOCC processes they model) rarely minimize delay, cancellations, and aircraft swaps simultaneously – they treat these as separable sub-problems solved and handed off in sequence, so a solution that is optimal at one stage can be infeasible once the next stage’s constraints are applied. Hu et al. [21] adds that the standard simplification in this literature – modeling delay cost as linear in delay minutes – does not match airline practice, where delay cost is nonlinear and airlines routinely trade mathematical optimality for decision speed. Kontodimou et al. [26]’s predictive turnaround-buffer model achieves a real, measured 37.1% reduction in propagated delay, but only by holding routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as external constraints – the improvement is genuine within its bounded scope, and the boundary itself is the risk this capstone investigates. Finally, Turhan, Gzara, and Elhedhli [47]’s recent (2026) study of the flight-dispatcher workload-balancing problem is notable for two reasons: it is one of only three studies the authors could find that treat dispatcher scheduling as an optimization problem at all, and it supplies a directly reusable formulation (a load-based objective, a Lagrangian-relaxation bound, and a metaheuristic) for treating human workload as a first-class, quantified constraint rather than an unmeasured externality – exactly the kind of factor Section 1.1 identifies as “harder to quantify” and therefore routinely omitted from today’s cost models.

Research gap

Individually, each of these tracks is well studied: sequential-versus- integrated optimization within airline planning, information-sharing protocols for turnaround, and disruption-recovery decision support each have mature, if incomplete, literatures, and precedents exist for deriving MBSE-based ATM architecture from an operational concept in other regions (GreAT, SESAR) and other aviation domains (UAM). What none of them do, individually or together, is connect these tracks into one traceable model. Each track studies its own slice of the trajectory-intent chain in isolation – planning *or* turnaround *or* dispatch *or* foreign-ATM architecture – using that slice’s own vocabulary and cost model, and each documents, in its own domain, the same underlying pattern: sequential or siloed decision-making that is locally rational and globally uncertain. No source in this register builds a single architecture that traces operational intent end-to-end – from an enterprise objective, through airline planning, day-of-operations control, and air traffic control, to the trajectory an aircraft actually flies – while keeping each stakeholder’s objectives, and the points where one stakeholder’s decision imposes a cost on another, explicit and inspectable in the same model. That is the gap this capstone’s reference architecture is intended to fill: not a new optimization method for any one of these tracks, but a structure that makes the couplings *between* them visible to a systems engineer or decision-support tool before a disruption forces them into view.

1.3 Stakeholders

Stakeholders were identified with a PESTLE (Political, Economic, Social, Technical, Legal, Environmental) sweep across the NAS, producing a register of several dozen stakeholder classes ranging from the FAA and ICAO down to local communities and future generations affected by long-run environmental cost. That full inventory is deliberately broader than any one model or optimization study can carry in detail, so it is narrowed, for the architecture and objective-ontology work, to seven categories that sit directly inside the trajectory-intent chain and actively trade against one another: **Airline**, **Passenger**, **ATC/ANSP**, **Airport**, **Flight Crew**, **Environmental/Societal**, and **Military**. Each of the seven has been carried through a per-category objective/cost/value analysis (classifying each objective as a hard constraint, an optimization objective, a cost/penalty, a measure of performance, or a measure of effectiveness, and rating whether it is directly, indirectly, or not coupled to aircraft trajectory) that is summarized here and used directly in Section 1.4 and Section 1.5:

- **Airline** – fuel cost, crew cost, aircraft utilization, maintenance, schedule integrity, passenger connections, delay cost, revenue, and dispatch reliability. Represented operationally by the AOCC, dispatch, and network-planning functions discussed in Section 1.2.
- **Passenger** – ticket price, travel time, connection reliability, schedule convenience, comfort, and disruption risk. A comparatively low-power stakeholder in the sense that passengers do not make operational decisions, but one whose travel-time and connection outcomes are a direct, measurable consequence of decisions made by every other stakeholder in this list.
- **ATC/ANSP** – safety and separation, sector workload, capacity, predictability, delay, and traffic complexity. Safety and separation are hard constraints that bound every other objective in this architecture; sector workload is the clearest example in this project of an externality (imposed by airline and individual-aircraft trajectory requests) that the imposing party does not itself pay for.

- **Airport** – gate utilization, runway utilization, surface congestion, turnaround performance, and passenger throughput. Sits at the A-CDM information-sharing boundary discussed in Section 1.2.
- **Flight Crew** – safety, workload, procedural compliance, schedule/duty constraints, and operational flexibility. The Captain’s authority to deviate from an assigned trajectory to preserve safety margin is this project’s clearest example of decision authority that can, and by design does, override every other stakeholder’s local objective.
- **Environmental/Societal** – CO₂, NO_x, contrail/climate effects, noise, local air quality, and community impacts. The lowest-power stakeholder category in this list – represented by advocacy groups, regulators, and “future generations” as a societal proxy rather than by an operational decision-maker – and the clearest example of a stakeholder whose costs are chronically externalized by every other category’s schedule-recovery and efficiency decisions.
- **Military** – mission effectiveness, mission timing, fuel availability, airspace access, operational security, and resilience. Included specifically at the civil-military airspace-access interface (special-use-airspace activation forcing civil rerouting), not as a detailed model of military mission planning – see Section 1.4.

Because this is a single-student capstone rather than a team project, there is no team-composition bias to check against; the mitigation instead was to seed the register broadly (the full PESTLE inventory) before narrowing to the seven working categories, so that low-power stakeholders were considered and then deliberately retained – Passenger and Environmental/Societal in particular – rather than dropped for convenience. The needs, resources, constraints, and information requirements underlying each category above are being developed into a full RACCI (Responsible, Accountable, Consulted, Informed, and a fifth authority dimension) matrix of NAS operational decisions; that matrix, and the deeper per-stakeholder detail, are in progress and will be reported in a subsequent submission.

1.4 Scope

The System of Interest (SOI) is the portion of the NAS that participates in the trajectory-intent chain for scheduled U.S. commercial (Part 121) air transportation: the organizations, systems, and information exchanges that carry intent from an airline’s strategic and network-planning objectives, through day-of-operations control and air traffic control, to the trajectory an aircraft actually flies and the guidance commands that execute it.

Included. Airline strategic/network planning and its sequentially-coupled sub-problems (schedule design, fleet assignment, aircraft routing, crew scheduling); day-of-operations functions (dispatch, the AOCC, weight and balance); turnaround and airport ground operations insofar as they exchange information across organizational boundaries (the A-CDM milestone model); air traffic control from clearance delivery through enroute, arrival, and approach control; and the onboard systems (FMS, autopilot, guidance) that translate a cleared trajectory into aircraft motion. The seven stakeholder categories in Section 1.3 are the SOI’s actor set.

Excluded. General aviation, all-cargo and charter operations outside scheduled Part 121 service, and international airspace beyond the NAS boundary are out of scope for direct modeling, though the SESAR/European and GreAT sources in Section 1.2 are used as comparative benchmarks, not as SOI content. Military aviation is excluded except at its one interface with the civil SOI: special-use-airspace activation and the civil rerouting it forces (Section 1.3); internal military

mission planning, weapons systems, and mission systems are treated as an opaque boundary condition. Urban Air Mobility and uncrewed systems are excluded from the SOI itself and used only as methodological precedent (Section 1.2). Below the aircraft/FMS interface – autopilot control-law design, control-surface actuation, and other airframe-internal behavior – is excluded; the architecture treats “aircraft motion” as the observable output of guidance commands, not as a flight-dynamics model.

Level of abstraction. The model spans enterprise objectives down to trajectory intent and the guidance commands that realize it, but not below that: the trajectory-intent chain (Strategic objective → mission plan → flight plan → ATC constraints → trajectory negotiation → FMS intent → guidance commands → aircraft motion, Section 1.1) is modeled at full fidelity end-to-end, while lower-level detail inside any one link – e.g., the specific numerical control law an FMS uses to track a cleared trajectory – is treated as a boundary input. Consistent with the per-objective abstraction judgments already made in the stakeholder objective/cost ontology (Section 1.3), an objective or decision is modeled at full trajectory fidelity when it is *directly* coupled to trajectory (e.g., ATC safety/separation, schedule-recovery delay cost, contrail avoidance) and treated as a coarser boundary condition when the coupling is indirect (e.g., ticket pricing, passenger throughput, crew duty-time constraints).

Boundary criteria. A candidate element is included as a *system* in the architecture if it produces, transforms, transmits, or consumes trajectory-relevant intent or information, and if a specific organization can be identified as holding authority or responsibility for it – consistent with the domain decomposition already adopted (governance, airspace management, airspace resources, flight operations, airport operations, aircraft systems, information services, infrastructure, and decision support; decision D-002). An element that merely exists in the aviation domain without a traceable connection to that chain (e.g., an airline’s internal HR system) is out of scope. A candidate *stakeholder* or actor is included if it holds authority over, is responsible for executing, or bears a measurable cost or benefit from a decision inside the trajectory-intent chain – the test applied when the seven-category working set in Section 1.3 was narrowed from the full PESTLE inventory.

Status. This is the project’s initial, working SOI boundary, consistent with the candidate scope already sketched in the project’s working notes; it closes the boundary-definition questions raised during research-framework setup (inclusion/exclusion, abstraction level, and system and stakeholder boundary criteria). It remains open to revision when alternative decompositions are compared against it later in the project (organization-based, lifecycle-based, physical, information-flow, and decision-authority views), and that comparison may refine or replace the domain decomposition without changing the SOI boundary stated here.

1.5 Project Deliverables

The project has two primary deliverables, developed together so that the second demonstrates the value of the first.

1. A SysML/Cameo reference architecture of the SOI defined in Section 1.4. This includes a NAS context diagram and stakeholder model; block definition diagrams (BDDs) for the domain decomposition; internal block diagrams (IBDs) for the critical interfaces identified along the trajectory-intent chain; an information-object and item-flow model; a nominal-flight activity diagram and responsibility swimlanes (grounded in the ConOps built from Section 1.2’s literature, especially the A-CDM milestone model and the AOCC/dispatch decision process); and a requirements model traced from stakeholder needs (Section 1.3) through requirements, systems, activities, and information exchanges, to the decisions that ultimately shape aircraft behavior.

2. A bounded optimization-demonstration capability built as one decision-support capability *inside* that architecture rather than as a standalone study – consistent with this project’s own framing (this is a systems-engineering capstone, not an optimization paper). Using the objective/cost ontology summarized in Section 1.3, the demonstration formulates one representative multi-stakeholder conflict (the leading candidate is airline fuel-cost minimization versus ATC/ANSP sector workload, identified in that ontology as the cleanest available decision variable with clean opposing measures on both sides), runs a single local-versus-system-level-optimum comparison and objective-weight sweep, and traces the resulting decision back through the architecture to show which stakeholders absorb which costs – the specific capability a standalone optimizer, decoupled from the architecture, could not show.

Success is assessed against measures that can be checked directly in the finished model and report, not by subjective judgment of “improvement”:

- **Stakeholder coverage** – all seven stakeholder categories (Section 1.3) represented in the architecture with an identified objective, decision authority, and at least one information dependency.
- **Traceability coverage** – a complete, unbroken chain from stakeholder need through requirement, system, activity, and information exchange, to a specific effect on aircraft trajectory or behavior, for at least the nominal-flight ConOps scenario.
- **Interface/requirement completeness** – all interfaces identified as critical in the IBD model are connected, and all requirements trace to at least one stakeholder need.
- **Demonstrated tradeoff** – the optimization capability produces at least one local-optimum-versus-SoS-optimum comparison that quantifies a cost externalized from one stakeholder onto another, and shows that externality changing as the objective weighting changes.
- **Architecture value-add** – an explicit statement, supported by the demonstration, of what the architecture exposes that the optimization result alone would not (i.e., which stakeholder absorbs the externality and through which interface).

1.6 Project Timing

The timeline below revisits the milestone schedule from the 503 Proposal submittal and maps each milestone to the corresponding section(s) of the project’s working to-do list (16 sections, research framework through final deliverables). Three planning activities (alternative decomposition exploration, myopic-optimization conflict analysis, and the capstone demonstration walkthrough) have no dedicated milestone in the original submittal; they are scheduled to fall between their logically adjacent milestones and are marked accordingly.

Target date	Milestone	Plan §	Status (as of September 14, 2026)
2026-09-07	PESTLE Analysis & System Definition (stakeholder discovery, SOI boundary, NAS context diagram)	S1, S7, S10	PESTLE discovery and stakeholder register complete; SOI boundary stated in Section 1.4

Target date	Milestone	Plan §	Status (as of September 14, 2026)
2026-09-10	Authority & Responsibility Models (RACCI, responsibility swimlanes)	S7, S10	Not started
2026-09-20	Interim Report #1 (this report)	S1–S4, S16	In progress – this document
2026-09-22	Sequence Diagrams (critical message flows)	S10	Not started
2026-10-01	Activity Diagrams (nominal-flight scenario walkthrough)	S10	Not started
2026-10-12	Requirement Traceability (needs → requirements → systems → activities → information)	S10	Not started
2026-10-17	Measures of Effectiveness (MOE/MOP hierarchy)	S8	Complete – see stakeholder objective/cost ontology
2026-10-20 (unscheduled, between 2026-10-17 and 2026-11-03)	Requirements Definition Alternative system decompositions compared against the domain decomposition	S10 S6	Not started Not started
2026-10-25	Internal Block Diagrams (critical interfaces connected)	S10	Not started
2026-10-29	Interface Definition (Interface Control Document)	S10	Not started
2026-11-03	Draft Architecture (BDDs & package structure)	S10	Not started
2026-11-08 (unscheduled, between MOE and Optimization Prototype)	Operational Scenarios (nominal-flight ConOps document) Myopic-optimization conflict analysis	S5 S9	Not started Not started
2026-11-17	Literature Review (annotated bibliography, 10–20 papers categorized)	S2–S4	In progress – 52 sources triaged, deep annotation ongoing
2026-11-21	Optimization Prototype (working code, unit tests)	S11, S12	Not started
2026-11-24	Integration of MBSE Architecture (traceability links, end-to-end scenario)	S10, S13	Not started

Target date	Milestone	Plan §	Status (as of September 14, 2026)
(unscheduled, between Integration and Final Paper)	Capstone demonstration walkthrough	S14	Not started
2026-12-18	Final Paper (complete draft)	S16	Not started
2026-12-20	Paper Submittal (official)	S16	Not started
2026-12-26	Final Baseline & Draft Paper (versioned model & draft paper)	S15	Not started

Two scheduling notes carry over from the original submittal and are flagged here rather than silently resolved: the submittal schedules the Literature Review milestone (2026-11-17) *after* most of the architecture milestones (2026-09-07 through 2026-11-03), later than this report’s own literature-review-before-architecture sequencing assumes – in practice, literature work is running continuously in the background (as reflected in Section 1.2) rather than strictly front-loaded. The Final Baseline & Draft Paper milestone (2026-12-26) is also dated *after* Paper Submittal (2026-12-20); this is being treated as an intentional post-submission archival/versioning step pending confirmation with the faculty adviser.

2 Assumptions and Methodology

Plan ref: to-do-list.md S1 — State the working assumptions carried through the project – see README.md “Working Assumptions” and `projects/nas-sos-capstone/index.md` “Working assumptions (guardrails)” (e.g., emphasis on structure/behavior/interfaces/traceability rather than optimization math; decision-support/optimization treated as a capability inside the architecture, not the whole subject).

Plan ref: to-do-list.md S6 — Describe the methodology for exploring and comparing alternative system decompositions (organization/authority-based, lifecycle/process-based, physical-system, information-flow, decision-authority, intent-to-trajectory) and the criteria used to evaluate them (what each exposes/obscures, where boundaries are ambiguous, which best supports optimization-impact analysis).

Plan ref: to-do-list.md S10 — Describe the MBSE methodology used to build the architecture in Cameo: package organization, diagram types produced (BDD, IBD, activity, sequence, requirements), and the traceability approach (stakeholder needs -> objectives -> requirements -> systems -> activities -> information exchanges -> aircraft behavior). Note per `workflows/update-architecture.md` that Cameo is the source of truth and the XML export is a generated artifact.

Plan ref: to-do-list.md S11 — Describe the optimization-study methodology: scenario selection criteria, system-state and decision-variable definitions, constraint definitions, the multi-objective formulation $J = w_1J_1 + w_2J_2 + \dots + w_nJ_n$, normalization approach for objectives with different units/scales, and the stakeholder weighting scheme. This section should be detailed enough that the scenario and formulation could be reproduced.

3 Results and Discussion

Plan ref: to-do-list.md S5 — Present the nominal-flight ConOps (strategic/commercial planning through day-of-operations, turnaround, flight execution, and postflight/continuation) as the operational backbone the architecture models. Summarize; put the full activity breakdown in Appendix A.1.

Plan ref: to-do-list.md S8 — Present the stakeholder objective/cost/value ontology: what "optimal" means per stakeholder (airline, passenger, ATC/ANSP, airport, flight crew, environmental/societal, military), and the classification of each objective as hard constraint, optimization objective, cost/penalty, MOE, or MOP. Highlight conflicting vs. aligned objectives and objectives with externalized costs.

Plan ref: to-do-list.md S9 — Present the myopic-optimization analysis: the definition of "local optimum" in the SoS context, candidate SoS measures of value, and the documented conflicts (decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence). See the flagged scope note below.

Scope note (flag, not a TODO to resolve silently): Project_To-Do List.md S9 catalogs eight candidate local-vs-SoS conflicts, each with eight documented attributes (64 cells). A 15–40 page report body cannot carry all eight in full; plan to present 2–4 illustrative conflicts in the main body (favoring whichever one becomes the S11–S14 demonstration scenario) and move the remaining catalog entries to Appendix A.4.

Plan ref: to-do-list.md S10 — Present the SysML architecture: context diagram, package structure, stakeholder model, BDDs, IBDs, information-object model, interface/item-flow model, nominal-flight activity diagram, responsibility swimlanes, and requirements model with traceability. Embed the most illustrative diagrams here; move the full diagram set to Appendix A.3.

Plan ref: to-do-list.md S12, S13 — Present the optimization/simulation results: the base-line scenario, single-stakeholder vs. balanced/SoS optimization outcomes, and how simulation entities/decisions/objectives/constraints trace back to SysML blocks, properties, goals, and requirements. Show how the architecture exposes impacts a standalone optimizer would omit (S13's key point).

Scope note: Project_To-Do List.md S12 as written (Pareto fronts, full objective-weight sweeps, sensitivity analysis, tipping-point identification across the whole experiment matrix) reads as a full optimization research study layered on top of the architecture work. Per this project's own working assumption ("not an optimization paper"; optimization is a capability inside the architecture, not the whole subject) and the 15–40 page report constraint, consider scoping S12's demonstration to one representative scenario with a single weight sweep and one Pareto-style comparison, rather than the full matrix – see the note added to `knowledge/questions/open-questions.md` under "Optimization study scope (S11)."

4 Recommendations

Plan ref: to-do-list.md S14, S15 (S16 bullet added this session) — Restate the main findings from Section 3 and provide recommendations for moving forward. Include the alternatives considered (e.g., alternative decompositions from S6, alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9's catalog).

NOTE: Project_To-Do List.md did not previously have a distinct "Recommendations" deliverable separate from "Conclusions and future work" – added to S16 this session so this section has an explicit home in the plan.

5 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its optimization-demonstration capability. Translate the S9 conflict analysis and S12 demonstration results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO₂, gate/runway utilization) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to S8’s “objectives whose costs are externalized onto another stakeholder” and S9’s conflict catalog).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template’s sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a “before” baseline in dollars, so a literal dollar-savings figure may not be available or meaningful. Project_To-Do List.md did not previously have a task that produces a quantified/monetized impact figure – an explicit “Impact / quantified benefits” bullet was added to S16 this session so at least one deliverable targets this section directly. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

6 Summary/Conclusions

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings/ recommendations, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., “reduces network-level delay propagation by an illustrative X% under the demonstrated weighting” rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the optimization demonstration (S12 scope note, Section 3), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

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A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 3).

A.2 Full Stakeholder Register and RACCI Matrix

Plan ref: to-do-list.md S7 — Complete stakeholder register (needs, goals, responsibilities, authorities, resources, constraints, information needs, measures of value) and the full RACCI matrix around operational decisions.

A.3 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in Section 3: full BDD/IBD set, requirements model, and complete traceability matrices (needs-¿objectives-¿requirements-¿systems-¿activities-¿information exchanges-¿aircraft behavior).

A.4 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.5 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 3.

A.6 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (evidence/source-register.md): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.